Qiskit Global Summer School 2026

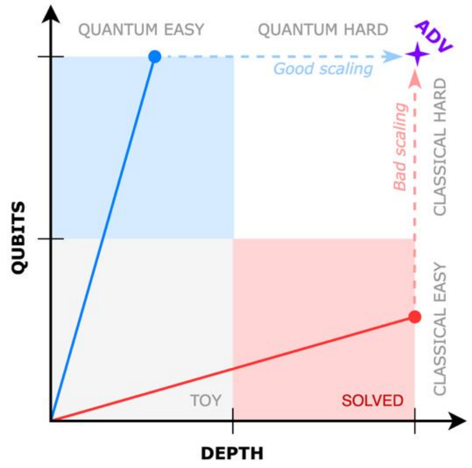
Lecture 3: Programming quantum computers

Ibrahim Shehzad
Quantum Algorithm Engineering
IBM Quantum

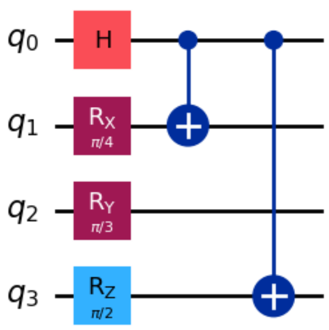
Overview

The Bloch sphere: a handy way to represent an arbitrary quantum state

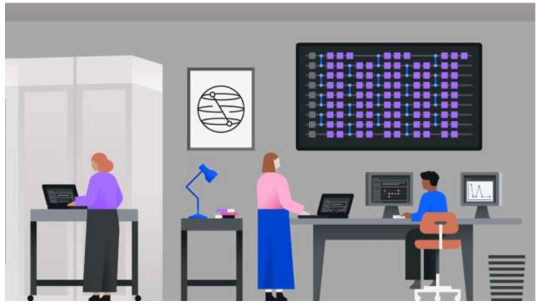
Shallow depth circuits and their utility for quantum



Universal quantum gate set: Single and two qubit gates



Hands-on quantum programming



Bloch sphere: A compact representation of a quantum state

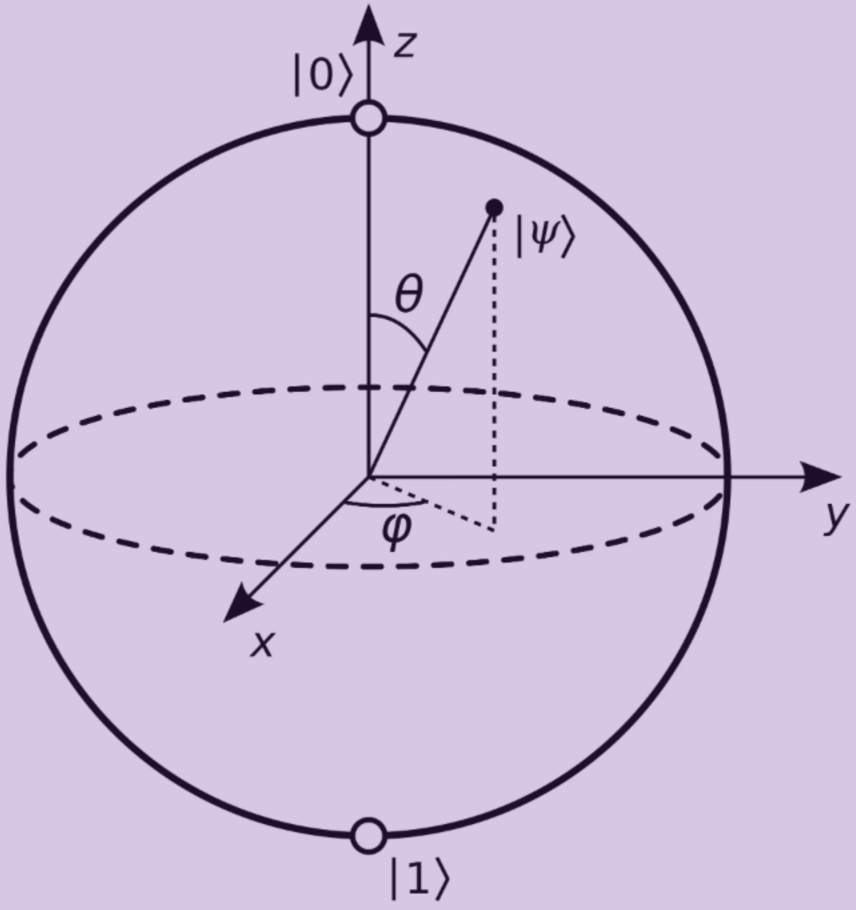
An arbitrary 2-state system (qubit) defined over the complex numbers has 4 degrees of freedom.

Normalization and modding out by the overall phase reduces this to 2, which is why an arbitrary quantum state can be represented on a unit 2-sphere (called the "Bloch" sphere).

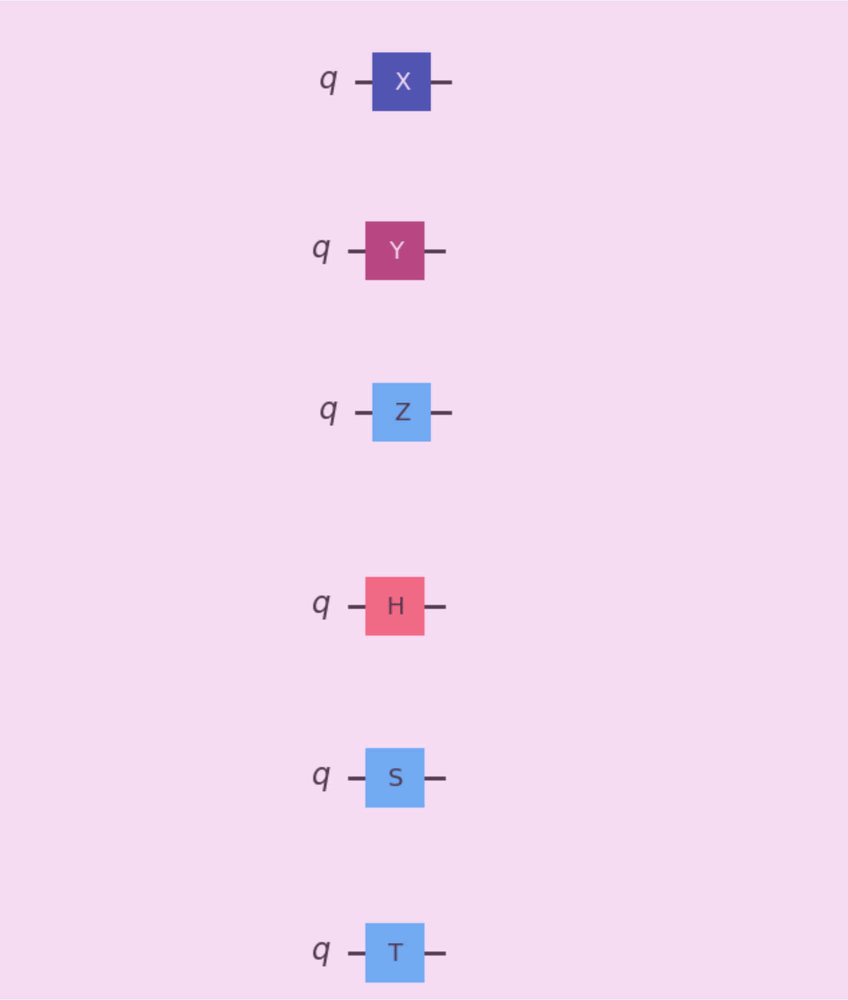
An arbitrary operation on a quantum state can be represented as a rotation on the Bloch sphere.

Pauli-X, Pauli-Y, Pauli-Z are generators of these rotations

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

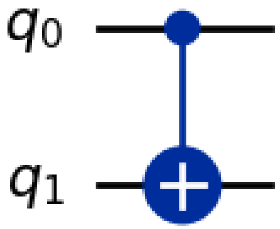


Towards a universal gate set: single qubit gates



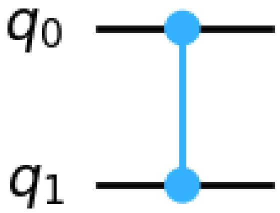
X (Bit flip) gate	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Y-gate	$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Z (Phase flip) gate	$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Hadamard gate	$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
S-gate	$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
T-gate	$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

Towards a universal gate set:
…but you need 2 qubit gates to create entanglement



$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Cx: The quantum version of the classical controlled bit-flip



$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Cz: The native 2-q gate on IBM’s Heron device

Universal quantum operations

An arbitrary quantum gate can be formed by composing two and one qubit gates.

An arbitrary two-qubit gate can be formed by composing a Cnot's with arbitrary rotations on the Bloch sphere

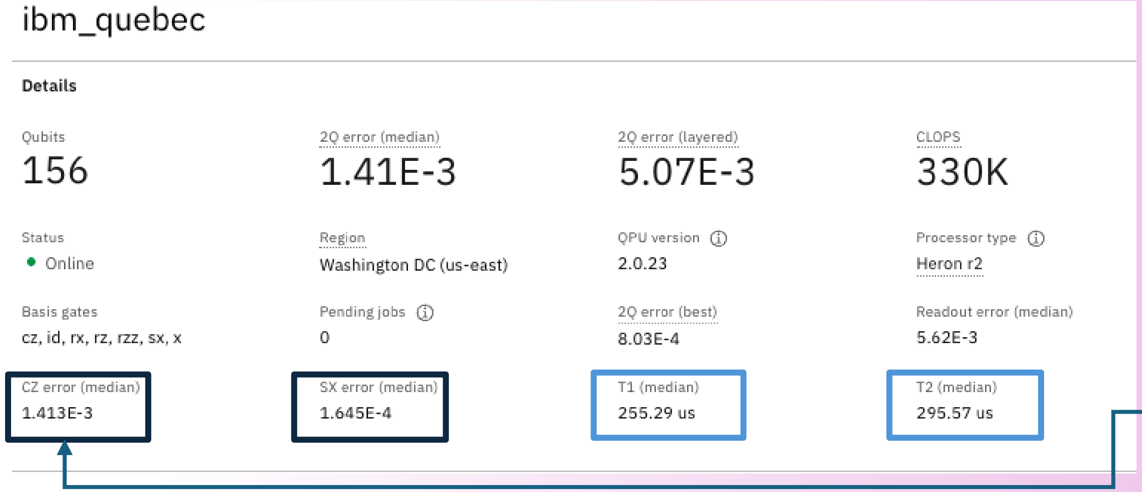
Another basis of universal gates: {Cx, T, H}

Important (non-universal) subgroup: The "Clifford" group: {H, S, Cx}

Gottesman-Knill theorem: Circuits comprising Clifford gates are efficiently classically simulable.

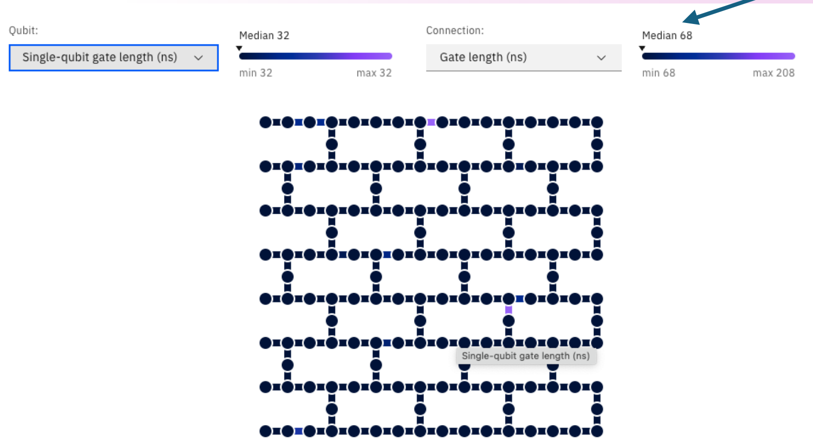
To really derive value from quantum executions, we need to focus on non-Clifford circuits

Large qubit-count and shallow-depth circuits suit the current hardware


ibm_quebec

Details

Qubits	2Q error (median)	2Q error (layered)	CLOPS
156	1.41E-3	5.07E-3	330K
Status	Region	QPU version	Processor type
Online	Washington DC (us-east)	2.0.23	Heron r2
Basis gates	Pending jobs	2Q error (best)	Readout error (median)
cz, id, rx, rz, rzz, sx, x	0	8.03E-4	5.62E-3
CZ error (median)	SX error (median)	T1 (median)	T2 (median)
1.413E-3	1.645E-4	255.29 us	295.57 us



"Useful" circuit sizes are limited by coherence times and gate errors.

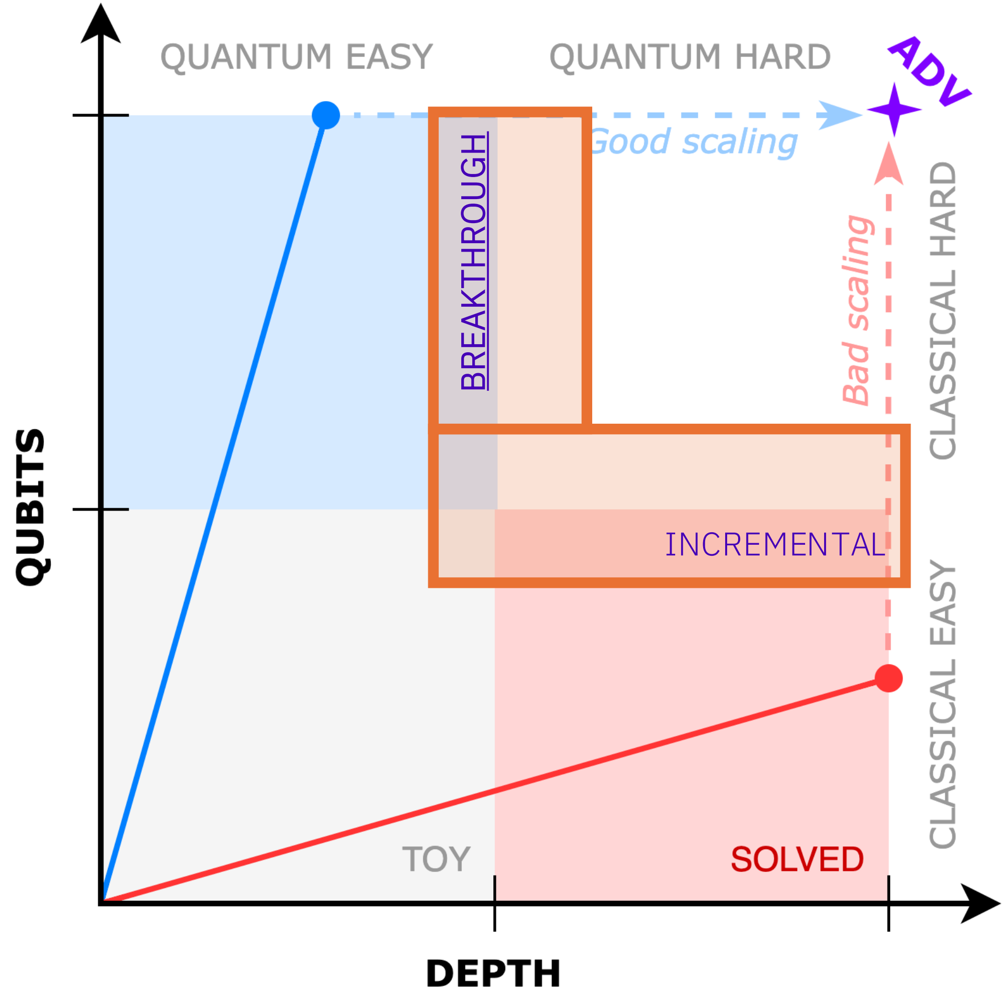
The median 2q error rate is an order of magnitude higher than the median 1q error rate. 2q gates also take longer to implement.

On pre-fault tolerant devices, this means that "shallow" circuits yield more meaningful results. On the newest IBM devices, this translates into ~ O(100) 2q gate layers, and ~10000 2q gates [which can be increased using post-selection, configuration recovery, error mitigation].

https://quantum.cloud.ibm.com/docs/en/guides/error-mitigation-and-suppression-techniques

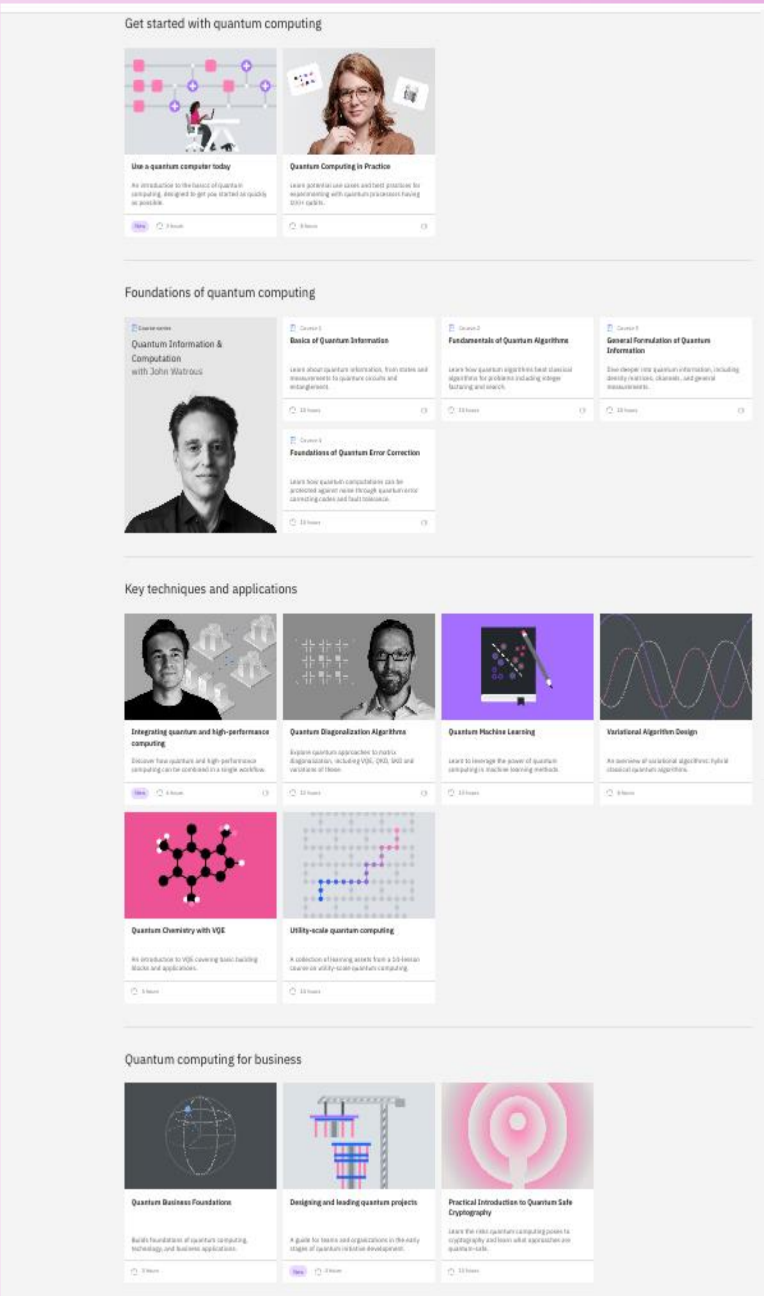
Scaling towards breakthroughs

- Classical computers are typically limited in memory (i.e. number of qubits) when trying to emulate quantum computational calculations.
- Non-fault-tolerant quantum computers are limited in the depth of the circuits that they can execute before noise becomes dominant.
- Therefore, useful quantum computing today lies in the medium depth and high qubit number regime.

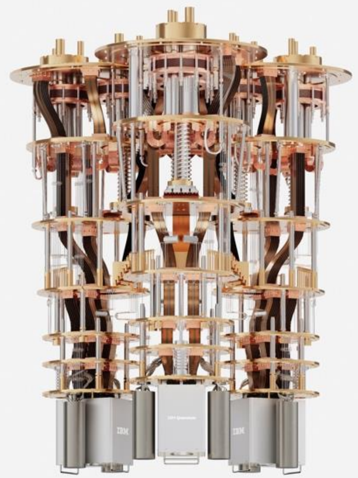


Further exploration:

A plethora of amazing courses on the IBM quantum platform.



In-depth tutorial on single-qubit gates in Qiskit:
https://github.com/Qiskit/textbook/blob/main/notebooks/ch-states/single-qubit-gates.ipynb



Next lecture:
Entanglement,
No-cloning theorem,
Much more…

IBM

IBM